

Does Heat Melt Ice??

(Analysing Arctic ice extents and temperature)

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Introduction

Every year in the Arctic, sea ice goes through a cycle of accumulating and receding. Daily satellite imagery has been used since 1978 to measure the extent of ice. In recent decades, the Arctic temperature has also been rising.

This project asked **how is the loss of Arctic sea ice correlated with Arctic temperature?** To get the answer, the trends of ice extent and temperature were visualized separately and then together to identify the strength of their correlation. This analysis used publicly available ice extent data from NSIDC[1] and temperature data from GISTEMP.[2]

Data Sample

The NSIDC dataset[1] contains the daily Arctic ice extents since October 1978, in million square kilometers. Arctic ice reaches its minimum in September,[3] so only data from rows with the month "09" were kept.

The GISTEMP dataset[2] is made up of an average temperature for every year in degrees Celsius relative to the 1951-1980 average. There are columns for specific geographical areas and latitudes. To focus specifically on the Arctic, only data from the column for latitudes 64N - 90N were analysed.

Both datasets were cleaned to remove rows with missing or invalid data and convert values into a usable format. The two sets of cleaned data were then combined by matching the years. The final data file used for analysis contains one row for each year, that year's average September ice extent (in km^2), the standard deviation of the daily values within that mean, and the temperature anomaly (in °C).

Analysis

The script to analyse the data was written in Python and uses numpy, pandas, script, and matplotlib. It includes self-tests that can be optionally run to test the functions for subsampling and slope calculations. The following were part of the analysis.

- **Trend lines:** Linear trends fitted to ice extent vs year and to temperature vs year. Slope of the trend line is determined per decade.
- **Slope statistical uncertainty:** The years were shuffled randomly (with a fixed seed), divided into 10 piles, and a slope fitted to each pile. The std of the 10 slopes is the statistical uncertainty of the main trend's slope.
- **Systematic uncertainty:** The slope is refitted over three different ranges to see if it depends on range. Half of the spread gives the systematic uncertainty.
- **Significance:** Slope is divided by its statistical uncertainty to get the number of sigma it is away from zero.
- **Correlation:** The Pearson correlation coefficient (r) was visualised by plotting ice extent against temperature. An r value close to -1 indicates warmer years are correlated to less ice.

Results and Discussion

Figure 1 shows how both the September ice extent and relative temperature have independently changed over time between 1979 and 2025. The ice shows a strong downward trend, while the temperature shows a strong upward trend. The fitted trends are:

$\Delta\text{Ice} = -0.76 \pm 0.31\ (\text{stat}) \pm 0.04\ (\text{syst})$ million km^2 per decade $(2.5\,\sigma)$

$\Delta\text{T} = +0.68 \pm 0.24\ (\text{stat}) \pm 0.06\ (\text{syst})$ °C per decade $(2.8\,\sigma)$

In the 46 years analysed, this shows an ice extent decrease of about 3.5 million km^2, which is a substantial loss from the roughly 7 million km^2 in 1979. The temperature increased by a little over 3°C. The figure also shows that the slope of the trend line doesn't change much when a different range of years is assessed.

Figure 2 shows the ice extent plotted against the relative temperature directly. The graph shows that the points follow a linear trend with a correlation coefficient of r = -0.89. This value for r supports a strong correlation between warmer years and less September ice. Dividing the trend of ice (Fig 1) by the trend of temperature (Fig 2), gives a rough sensitivity of 1.1 million km^2 of ice lost per °C of warming.

There are some limitations with this project. For instance, while the September ice data is daily measurements, the temperature data is in the form of annual means and doesn't include uncertainty values. Ideally, daily temperature readings for every September would be compared against daily ice measurements, but I was unable to find such data that was publicly accessible.

Finally, it is important to remember that correlation does not imply causation. Even though concluding that higher temperatures cause ice to melt may seem intuitive, the Pearson correlation simply shows that both trends are changing linearly with time. Additionally, large phenomena like Arctic ice melt can rarely be traced to a single cause, and factors like ocean currents, wind, ice thickness, and human interference must also be examined.

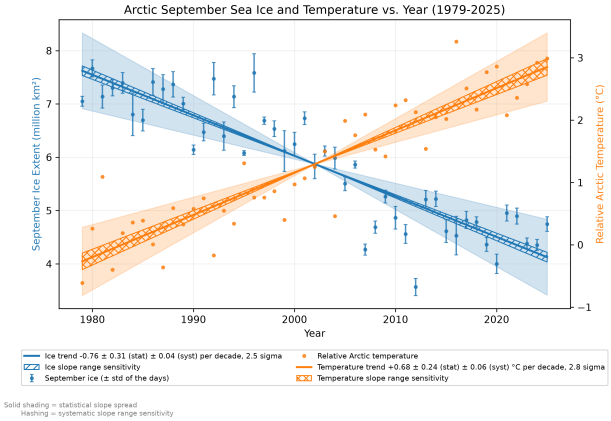


Figure 1: September Arctic sea ice extent (blue, error bars show the std of the daily values) and Arctic temperature anomaly (orange) plotted against the year. Shaded areas to show slope's statistical uncertainty, hashed areas show systematic uncertainty of the slope.

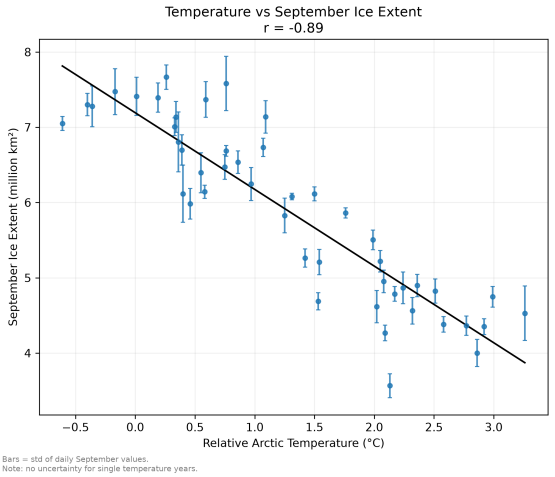


Figure 2: September average ice extent vs. Arctic temperature (relative to 1951-80 average). One point per year 1979-2025. Linear fit shown as visual guide for Pearson correlation, r = -0.89.

Conclusion

This analysis showed that Arctic sea ice in September has declined by 0.76 +/- 0.31 (stat) +/- 0.04 (syst) million km^2 per decade since 1979, and the Arctic has warmed by 0.68 +/- 0.24 (stat) +/- 0.06 (syst) °C per decade. These two variables have a strong anti-correlation of r = -0.89, showing that they could be related. However, this alone cannot prove or disprove that one causes the other.

References

[1] Fetterer, F., Knowles, K., Meier, W. N., Savoie, M., Windnagel, A. K., and Stafford, T. (2025). *Sea Ice Index. (G02135, Version 4)*. [Data Set]. Boulder, Colorado USA. National Snow and Ice Data Center. DOI: 10.7265/a98x-0f50. `https://nsidc.org/data/g02135/versions/4`.

[2] GISTEMP Team, 2026. *GISS Surface Temperature Analysis (GISTEMP), version 4*. NASA Goddard Institute for Space Studies. https://data.giss.nasa.gov/gistemp/.

[3] NSIDC. (2020). *How is Arctic sea ice changing?* Ask a Scientist, National Snow and Ice Data Center. https://nsidc.org/learn/ask-scientist/how-arctic-sea-ice-changing#anchor-references.